Deployment

> Part of the React curriculum. This document covers static hosting, containerized deployment, CI/CD for a frontend build, service-worker caching and edge-gated routes, and the special case where the build itself is the release.

1. The build produces static files

Unlike the Python API (Python API Backend, Deployment), a React app built with a tool like Vite compiles down to a set of **static files** — HTML, JavaScript, CSS, and assets — with no server-side runtime of its own required to serve them:

```
npm run build
# produces a dist/ directory of static files
```

This is a meaningfully simpler deployment shape than a backend service: there's no process to keep running, no server-side crash to recover from — just files that need to be served over HTTP, which almost any static file host or CDN can do.

2. Static hosting platforms

Option	What it is
Netlify / Vercel	Purpose-built static-site/frontend hosting — connect a Git repository, and the platform builds and deploys on every push automatically, including preview deployments per pull request
Cloud object storage + CDN (S3 + CloudFront, Google Cloud Storage + Cloud CDN)	Upload the built files to object storage, serve them through a CDN in front of it for caching and global distribution
A plain web server (nginx, serving files directly)	Simplest possible option if you're already running your own infrastructure

For most projects, a purpose-built platform (Netlify/Vercel) is the least-effort option and handles CDN distribution, HTTPS, and build automation for you; object storage + CDN is the more manual, more configurable alternative common in larger cloud-native deployments.

3. Client-side routing needs a server-side fallback rule

Since routing (§1 of Routing & API Integration) happens entirely in the browser via JavaScript, the *server* needs to be configured to serve the app's single `index.html` for **any** path the app defines a route for — not just the literal root path. Without this, directly loading (or refreshing) a URL like `/detail/123` would 404 at the server level, since

no actual file named `detail/123` exists; only client-side JavaScript (which hasn't loaded yet, because the server never returned `index.html` in the first place) knows how to render that route.

```
location / {
    try_files $uri /index.html;
}
```

Most static hosting platforms (Netlify, Vercel) handle this automatically for a detected SPA; configuring it yourself (as above) is necessary if you're serving the build from a plain nginx server or a generic object-storage + CDN setup.

4. Containerized deployment

If you'd rather deploy the built static files as part of a container (e.g., to keep it consistent with how a colocated backend is deployed), a common pattern is a two-stage Dockerfile: build the app in one stage, then copy only the built static output into a minimal nginx image:

```
FROM node:20 AS build
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npm run build

FROM nginx:alpine
COPY --from=build /app/dist /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
```

This **multi-stage build** keeps the final image small — it contains only nginx and the built static files, not the entire Node.js toolchain and `node_modules` needed to produce them.

5. Environment configuration at build time vs. deploy time

Recall from Routing & API Integration, §8.2 that Vite bakes environment variables in at **build time**, not read fresh at container/process start the way a backend service typically does. This means "one build, configured per environment via env vars at deploy time" (the ideal from the architecture overview, §2.7) doesn't apply quite as directly to a frontend build as it does to a backend — if a value genuinely needs to differ between environments, you either produce a separate build per environment, or read the value at runtime instead of build time (e.g., fetching a small runtime config file the app loads on startup, rather than baking the value into the JavaScript bundle itself).

6. CI/CD

A typical pipeline for a frontend build:

1. **On every push**: run the test suite (Testing), a linter, and a type check (`tsc --noEmit`).
2. **On merge to main** (or a release trigger): run `npm run build`, producing the static output.
3. **Deploy**: upload the built files to the hosting platform/CDN, or build and push a container image if using the containerized approach (§4).

As with the backend (Python API Backend, Deployment, §6), the key property to preserve is that nothing reaches production without first passing the automated test suite and type check.

7. Progressive web apps and edge-gated routes

A service worker (generated by a plugin such as Workbox) makes the app installable and fast on repeat visits, and introduces a cache you must now reason about on every release.

- **Pick a strategy per kind of request.** Built, content-hashed JS/CSS/icons: precache and serve from the cache. API calls: **network-first with a short timeout** (a few seconds) that falls back to a cached copy only when the network is down — for live data, freshness matters more than offline access. Choose `registerType: 'autoUpdate'` so a new worker takes over without asking.
- **A navigation fallback needs a deny-list.** The service worker answers a top-level navigation to any unknown path with the cached app shell (that is how client-side routing works offline). Two classes of path must be **excluded** from that fallback, or the worker will answer them with the SPA instead of letting the request reach the server:
    1. any path with a **file extension** (a direct link to a downloadable `.pptx`/`.pdf` should download, not render `index.html`);
    2. any path the **backend or edge owns** — API prefixes, and especially routes gated by an identity-provider login. A gated page served from the cached shell never touches the edge, so an expired session is *masked*: the user sees a stale page whose API calls then fail. Build the deny-list in one tested function and extend it for every new gated route. The pattern in the generated worker must be a regex literal, not a function closing over a build-time variable — the worker is serialised to a standalone file where that variable doesn't exist.
- **The worker script itself must not be cached.** Serve `sw.js` with `no-cache` (or `must-revalidate`), including from any CDN or edge rule in front of it; a worker cached for a day delays every release by a day.
- **Verifying a deploy needs a clean browser state.** A browser profile that has already visited keeps its registered worker, so "open a new tab" is not enough to see a new release. Unregister the worker (DevTools → Application) and reload with a cache-busting query, or use a fresh profile; in an automated check, assert the content-hashed asset filename the page loads.
- **Gated fetches need the identity provider in the page's CSP.** A page's own `fetch()` to an endpoint behind an identity-provider login follows a redirect to the provider's domain; unless that domain is allowed in the content-security `connect-src`, the redirect fails silently and the call looks like a network error.

8. When the build *is* the release

If the production server simply serves the build directory from disk (a static preview server, for instance), then **running the build command is itself the point of no return**: the new files are live the moment they are written, before any restart or confirmation. Consequences:

- Build only when you are ready to ship, and treat the build as the deploy step in the checklist, not as a preparatory one. Keep the previous output (or be able to rebuild a known commit quickly) as the rollback.
- If a build-time variable such as the base path is baked into the output, the build must be run **with the production value**; restarting the server never fixes a wrong one.
- If a shared design system is consumed from **source** by path alias (no published package), get its checkout to the commit the app pins *first* — installing its dependencies and regenerating its (uncommitted) token files, which a setup script can do in one step — then pull and build the app. The app's build reads the dependency's current source, so building against a stale checkout, or one with missing generated tokens, fails or ships the wrong components.
- Editing a service definition's environment usually needs the service manager to *unload and reload* the job, not just restart it; a restart re-runs the already-loaded definition and silently keeps the old environment.
- Confirm with evidence, not feel: the content-hashed bundle filename the live page requests is the reliable proof that the fresh build, rather than a cached or stale one, is serving.

9. Core tools summary

Tool	Used for
Vite's `build` command	producing the deployable static output
Netlify / Vercel	zero-infrastructure static hosting with automatic Git-based deploys
nginx	serving the built files yourself, with the SPA fallback rule (§3)
Docker (multi-stage build)	packaging the build alongside a colocated backend's own deployment
vite-plugin-pwa (Workbox)	service worker generation, precache and runtime caching rules (§7)

See also

- React curriculum index
- Testing — verifying the app before it's deployed
- Architecture Overview, §11 for how this fits alongside the API's own deployment